

Analysis of Craniofacial Growth Patterns Using Linear Mixed-Effects Models

Using longitudinal data on craniofacial development, this note identifies clear age-related increases in the distance between the pituitary and the pterygomaxillary fissure, with growth rates averaging 0.66 mm per year. Males display higher variability and faster growth, while females show more consistent patterns. Linear mixed-effects models confirm the presence of individual- and gender-level differences, with the preferred specification allowing both intercepts and slopes to vary. These findings support the use of flexible modeling frameworks for analyzing heterogeneous growth trajectories.

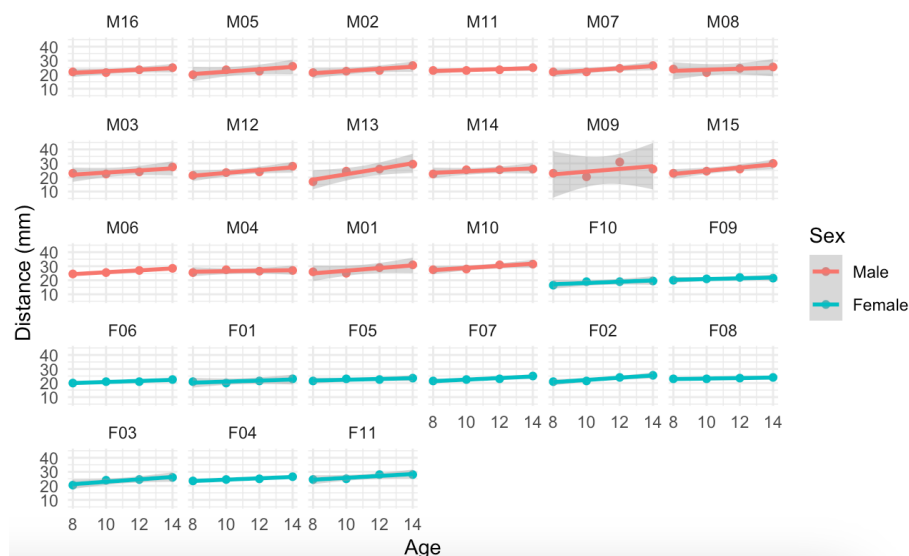
I. Introduction

This note analyzes craniofacial growth trajectories, focusing on changes in the distance between the pituitary and the pterygomaxillary fissure over time. The analysis is based on the Orthodont dataset from the *nlme* package in R, which provides longitudinal measurements of craniofacial development during childhood and adolescence. By leveraging this dataset, the study seeks to quantify the relationship between age and craniofacial growth and to examine whether systematic differences exist between males and females, as reflected in baseline levels and growth rates. In this context, baseline differences are captured by the intercepts of the models, while growth rates correspond to the slopes associated with age.

II. Exploratory Analysis

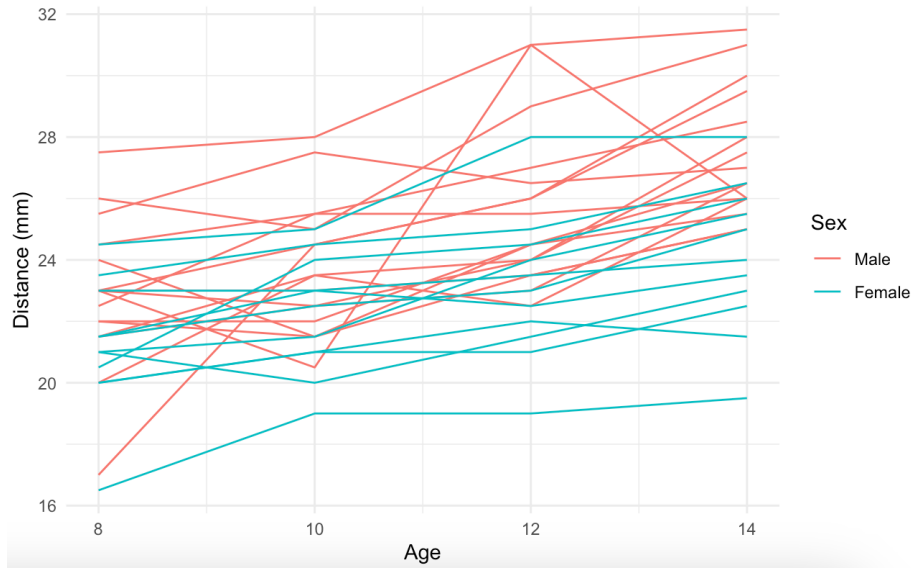
To establish an initial understanding of the data, individual growth profiles were graphed by subject and disaggregated by gender (Figure 1). The results show a consistent upward trend in the distance between the pituitary and pterygomaxillary fissure as age increases, aligning with expected natural growth patterns between ages 8 and 14. Profiles for males were generally positioned at higher levels compared to females, suggesting systematic differences in craniofacial development between genders. These patterns imply that, even from early ages, males tend to have greater craniofacial distances.

Figure 1: Individual Craniofacial Growth Profiles by Subject and Gender



Further examination, through overlaying individual growth trajectories by gender, revealed that males tended to exhibit greater variability in their craniofacial growth patterns compared to females (Figure 2). While both groups demonstrated a consistent upward trend with age, the trajectories for males displayed a wider range of initial distances and growth paths. In contrast, the trajectories for females appeared more tightly clustered, indicating relatively greater uniformity in both starting levels and growth rates.

Figure 2: Overlaid Craniofacial Growth Trajectories by Gender



Based on these initial observations, the use of linear mixed-effects models was considered appropriate to formally assess differences in growth patterns between groups.

III. Model Estimation

Building on the exploratory analysis, two linear mixed-effects models were estimated to formally evaluate the relationship between age and craniofacial distance.

- **Model 1:** Incorporated a random intercept for each subject, allowing individuals to have distinct baseline distances while assuming a common growth rate across subjects. The model can be expressed as:

$$distance_{ij} = \beta_0 + b_{i0} + \beta_1 age_{ij} + \varepsilon_{ij}$$

$$where\ b_{i0} \sim N(0, \sigma_{b_0}^2) \text{ and } \varepsilon_{ij} \sim N(0, \sigma^2)$$

- **Model 2:** Extended Model 1 by introducing a random slope for age, allowing both the intercept and the growth rate to vary across individuals. The model is specified as:

$$distance_{ij} = \beta_0 + b_{i0} + (\beta_1 + b_{i1})age_{ij} + \varepsilon_{ij}$$

$$where\ b_{i0} \sim N(0, \sigma_{b_0}^2), b_{i1} \sim N(0, \sigma_{b_1}^2), and\ \varepsilon_{ij} \sim N(0, \sigma^2)$$

Both models confirmed a positive and statistically significant relationship between age and craniofacial distance ($p < 0.001$), with an estimated slope of approximately 0.66 mm per year.

The goodness-of-fit criteria are presented in Table 3.

Table 3: Goodness-of-Fit Criteria

Criterion	Model 1 (Random Intercept)	Model 2 (Random Intercept and Slope)
AIC	455.00	454.64
BIC	465.66	470.62
Log-Likelihood	-223.50	-221.31

While Model 2 achieved a slightly lower AIC and a higher log-likelihood, indicating a marginally better fit, the BIC slightly favored Model 1. However, the differences were not substantial.

Model 1 Estimates (Annex 1):

- Random intercept standard deviation: 2.11 mm.
- Residual standard deviation: 1.43 mm.
- Correlation between intercept and slope: -0.845.

Model 1 assumes a common growth rate across individuals, thereby limiting its ability to capture variations in growth dynamics.

Model 2 Estimates (Annex 2):

- Random intercept standard deviation: 2.33 mm.
- Random slope standard deviation: 0.23 mm/year.
- Residual standard deviation: 1.31 mm.
- Correlation between intercept and slope: -0.848.

The inclusion of a random slope in Model 2 allowed for a better representation of variations in individual growth trajectories. In particular, the negative correlation between intercept and slope suggests that individuals with higher initial distances tend to grow more slowly.

Given its greater flexibility in capturing heterogeneity in growth patterns, Model 2 was selected as the preferred specification for further analysis.

IV. Interpretation of Fixed and Random Effects

Analysis of the fixed effects indicated that the intercept was estimated at 16.76 mm, corresponding to the theoretical distance at age zero. Given that the sample included only individuals aged 8 years and older, the intercept should be interpreted cautiously. Each additional year of age was associated with an average increase of 0.66 mm in craniofacial distance (p-value < 0.001).

The random effects analysis revealed substantial variability across individuals in both baseline measurements and growth rates. A negative correlation (-0.848) between intercept and slope further suggests that individuals starting with higher distances tend to grow more slowly, while those with lower initial distances tend to experience more rapid growth.

V. Gender-Specific Models

To better capture gender-specific growth dynamics, separate models were estimated for males and females, maintaining the random intercept and slope structure.

Males (Annex 3):

- Average intercept: 16.34 mm.
- Average slope: 0.78 mm/year.
- Greater variability in baseline levels (standard deviation of intercept: 2.67 mm) and growth rates (standard deviation of slope: 0.19 mm/year).
- Strong negative correlation between intercept and slope (-0.79).

Females (Annex 4):

- Average intercept: 17.37 mm.
- Average slope: 0.48 mm/year.
- Lower variability in baseline levels (standard deviation of intercept: 1.88 mm) and growth rates (standard deviation of slope: 0.16 mm/year).
- Moderate negative correlation between intercept and slope (-0.35).

These gender-specific results highlight distinct growth patterns: males demonstrated faster growth rates despite slightly lower initial distances, whereas females exhibited higher baseline distances but more moderate growth rates. Greater heterogeneity was also observed among males, both in initial measurements and in growth trajectories.

VI. Conclusion

The analysis underscores significant gender-based differences in craniofacial growth during childhood and adolescence. Males exhibited faster growth rates and greater individual variability compared to females, who showed higher initial measurements and more uniform growth patterns.

These findings reinforce the importance of applying flexible modeling approaches—such as linear mixed-effects models with random intercepts and slopes—to accurately capture individual and group-level growth dynamics. Such methodologies are crucial when analyzing longitudinal biomedical data where heterogeneity across individuals and subgroups is expected.

Annex 1. Model 1 Detailed Results

Linear mixed-effects model fit by REML

Data: Orthodont

AIC	BIC	logLik
455.003	465.656	-223.501

Random effects:

Formula: ~1 | Subject

	(Intercept)	Residual
StdDev:	2.115	1.432

Fixed effects: distance ~ age

	Value	Std.Error	DF	t-value	p-value
(Intercept)	16.761	0.802	80.000	20.889	0.000
age	0.660	0.062	80.000	10.716	0.000

Correlation:

(Intr)

age	-0.845
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Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-3.665	-0.535	-0.013	0.487	3.722

Number of Observations: 108

Number of Groups: 27

Annex 2. Model 2 Detailed Results

Linear mixed-effects model fit by REML

Data: Orthodont

AIC	BIC	logLik
454.637	470.617	-221.318

Random effects:

Formula: ~age | Subject

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	2.327	(Intr)
age	0.226	-0.609
Residual	1.310	

Fixed effects: distance ~ age

	Value	Std.Error	DF	t-value	p-value
(Intercept)	16.761	0.775	80.000	21.620	0.000
age	0.660	0.071	80.000	9.265	0.000

Correlation:

(Intr)

age	-0.848
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Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-3.223	-0.494	0.007	0.472	3.916

Number of Observations: 108

Number of Groups: 27

Annex 3: Male-Specific Model Results

Linear mixed-effects model fit by REML

Data: Orthodont_male

AIC	BIC	logLik
285.137	297.900	-136.569

Random effects:

Formula: ~age | Subject

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	2.667	(Intr)
age	0.189	-0.788
Residual	1.609	

Fixed effects: distance ~ age

	Value	Std.Error	DF	t-value	p-value
(Intercept)	16.341	1.210	47.000	13.506	0.000
age	0.784	0.102	47.000	7.722	0.000

Correlation:

(Intr)

age	-0.926
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Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-2.655	-0.594	0.019	0.576	3.158

Number of Observations: 64

Number of Groups: 16

Annex 4: Female-Specific Model Results

Linear mixed-effects model fit by REML

Data: Orthodont_female

AIC	BIC	logLik
149.429	159.855	-68.714

Random effects:

Formula: ~age | Subject

Structure: General positive-definite, Log-Cholesky parametrization

	StdDev	Corr
(Intercept)	1.884	(Intr)
age	0.161	-0.354
Residual	0.668	

Fixed effects: distance ~ age

	Value	Std.Error	DF	t-value	p-value
(Intercept)	17.373	0.761	32.000	22.841	0.000
age	0.480	0.066	32.000	7.242	0.000

Correlation:

(Intr)

age	-0.637
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Standardized Within-Group Residuals:

Min	Q1	Med	Q3	Max
-1.854	-0.468	0.068	0.430	1.592

Number of Observations: 44

Number of Groups: 11